Fiscal Year:	FY 2007	**Task Last Updated:**	FY 06/27/2007
PI Name:	Meiners, Jens-christian Ph.D.		
Project Title:	Microfluidic Single-Molecule Biosensor		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	48109-1120	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-04
Start Date:	06/01/2004	**End Date:**	12/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE until 12/31/2006, per A. Chu 6/23/06		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Walter, Nils (University of Michigan)		
Grant/Contract No.:	NNA04CD01G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The ideal biosensor is extremely sensitive, specific for the target of interest, adaptable to a wide range of target analytes, and, in particular for spaceflight applications, compact, rugged, and consumes very little energy or other resources. We propose to build and characterize a prototype biosensor based on catalytic RNA molecules on a microfluidic platform that can detect single analyte molecules with high specificity. RNA aptamers can recognize a wide variety of target molecules, such as cytokines or radiation-damaged DNA nucleotides with exquisite specificity. Target recognition is reported through the cleavage of a reporter RNA molecule, which in turn causes a change in fluorescence of attached fluorophores. This change can be detected optically down to the single-molecule level. The biosensor will be constructed on a microfluidic platform that contains all provisions for sample- and reagent handling and regeneration of sensor and reporter molecules on a single, robust and easy-to handle chip. It is manufactured in a planar multilayer soft lithography process that allows the incorporation of active elements such as valves and pumps on the chip. The chip is		

	packaged in a transparent epoxy casing to provide mechanical stability and robust interconnects. The sensor molecule itself will be anchored to a hybridization spot, allowing the in-situ assembly and regeneration of the active part of the sensor. The sensor chip will be characterized with respect to sensitivity, specificity, and reliability using a theophylline analyte, paving the way for more complex analytical applications, in which a suite of different sensor molecules establishes a comprehensive analytical profile of the sample.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of novel biosensors with extremely high sensitivity and specificity is important for a variety of biological and biomedical applications, ranging from the search for life in space to medical diagnostics. The developed catalytic RNA biosensor is a promising candidate for such applications.
Task Progress:	We have developed the novel biochemistry required for biosensing with catalytic RNA molecules, the single-molecule read-out using fluorescence resonant energy transfer, the microarray technology for anchoring the sensor substrate molecules, and a microfluidic platform to turn this sensor into a robust device.
Bibliography Type:	Description: (Last Updated: 06/26/2007)
Articles in Peer-reviewed Journals	Blumberg S, Gajraj A, Pennington MW, Meiners JC. "Three-dimensional characterization of tethered microspheres by total internal reflection fluorescence microscopy." Biophys J. 2005 Aug;89(2):1272-81. Epub 2005 May 27. PMID: 15923224 , Aug-2005
Articles in Peer-reviewed Journals	Newby Lambert M, Vocker E, Blumberg S, Redemann S, Gajraj A, Meiners JC, Walter NG. "Mg2+-induced compaction of single RNA molecules monitored by tethered particle microscopy." Biophys J. 2006 May 15;90(10):3672-85. Epub 2006 Feb 24. PMID: 16500956 , May-2006